PATHFINDER: AN INDOOR NAVIGATION WEB APPLICATION FOR THE PHYSICAL SCIENCES BUILDING IN THE UNIVERSITY OF THE PHILIPPINES LOS BAÑOS

DAVE CHRISTIAN P. PERALTA

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The special problem attached hereto, entitled "**PATHFINDER: AN INDOOR NAVIGATION WEB APPLICATION FOR THE PHYSICAL SCIENCES BUILDING IN THE UNIVERSITY OF THE PHILIPPINES LOS BAÑOS**," prepared and submitted by **DAVE CHRISTIAN P. PERALTA** in partial fulfillment of the requirements for the degree of **BACHELOR OF SCIENCE IN COMPUTER SCIENCE**, is hereby accepted.

DR. JUAN DELA CRUZ
Member, Guidance Committee

_____
Date Signed

DR. MARIA SANTOS
Member, Guidance Committee

_____
Date Signed

ARIAN J. JACILDO
Chair, Guidance Committee

_____
Date Signed

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science.

DR. PABLO OCAMPO
Director, Institute of Computer Science

_____
Date Signed

DR. ANNA REYES
Dean, Graduate School

_____
Date Signed

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LIST OF ABBREVIATIONS

2D	Two-dimensional
3D	Three-dimensional
API	Application programming interface
AR	Augmented reality
AWS	Amazon Web Services
CAD	Computer-aided design
CORS	Cross-Origin Resource Sharing
EC2	Elastic Compute Cloud
ERD	Entity relationship diagram
ETA	Estimated time of arrival
GIS	Geographic information system
GPS	Global Positioning System
IMM	Indoor mapping and modeling
IPMN	Indoor positioning, mapping, and navigation
IPS	Indoor positioning system
LBS	Location-based services
PWD	Persons with disability
RDS	Relational Database Service
RFID	Radio frequency identification
SUS	System Usability Scale
SVG	Scalable Vector Graphics
UPLB	University of the Philippines Los Baños
UPMO	University Planning and Maintenance Office
UWB	Ultra-wideband

ABSTRACT

PERALTA, DAVE CHRISTIAN P. Institute of Computer Science, College of Arts and Sciences, University of the Philippines Los Baños. September 2026. **PathFinder: An Indoor Navigation Web Application for the Physical Sciences Building in the University of the Philippines Los Baños.**

Adviser: Fermin Roberto G. Lapitan

Indoor wayfinding remains challenging in multistory buildings, where satellite-based positioning is unreliable and indoor positioning systems require costly, hardware-dependent infrastructure that is difficult to scale across floors and buildings. This study developed and evaluated a web-based indoor navigation application for the Physical Sciences Building at the University of the Philippines Los Baños, designed as a scalable and software-driven alternative to hardware-based positioning and static physical floor plans. The application provides a navigation interface that guides users between selected origin and destination points, and a layout builder that allows building administrators to create and publish digital map layouts for display within the application. Usability was evaluated with two participant groups: six regular users, who assessed navigation accuracy through a standardized usability questionnaire and open-ended questions, and four building administrators, who assessed the layout builder's functionality qualitatively. Users reported a mean usability score of 81.67 out of 100, rated "Excellent," with positive feedback on route clarity, while administrators found the layout builder intuitive but identified specific areas for refinement. These findings support the application as a viable, scalable approach to indoor navigation.

Keywords: indoor navigation, wayfinding, layout builder, web application

CHAPTER I

INTRODUCTION

1.1 Background of the Study

The world is a huge place, and we may often find ourselves getting lost in unfamiliar places. Throughout history, we have invented various tools, starting with the compass and the map, to help us with navigation. Nowadays, navigation is made easy thanks to modern tracking and mapping technology. In the Philippines, applications like Waze help to ease the transportation crisis [1].

One of the most commonly used technologies is the Global Positioning System (GPS), a publicly available mapping system created by the United States government that uses satellites to approximate the location of a GPS receiver contained in a device such as a laptop or a mobile phone. This allows users to obtain directions to and from a particular place, and it is especially useful in a travel setting. Popular applications that use GPS for navigation include Google Maps, Waze, and Apple Maps.

However, due to its top-down view from the satellites, GPS technology works best outdoors. It does not work well for buildings with multiple levels. To solve this, we can use an indoor positioning system (IPS), which works like a miniature version of the GPS. Specifically, we can install specialized tracking devices in various places inside the building. Several technologies can be used as positioning methods for the IPS, some of which include Wi-Fi, infrared, Bluetooth, and computer vision.

To further improve the IPS in terms of visualization, we can implement an indoor map where we can mark the locations of each landmark, as well as the location of the user. These maps can be derived from the floor plans of the building, and they can be

implemented without an IPS. Examples of applications that allow us to design indoor maps are Mappedin and ArcGIS Indoors. Google Maps also supports indoor maps, but only a few buildings currently have this functionality.

1.2 Statement of the Problem

Globally, hundreds of thousands of buildings are being constructed every year. In the Philippines, the total number of buildings constructed in the year 2022 was 169,432, which was 4.2 percent higher than the 162,619 constructions in 2021 [2]. Since GPS is not an effective solution for mapping the inside of these buildings, there is a growing need for indoor mapping solutions.

The Physical Sciences Building on the University of the Philippines Los Baños campus is one of these buildings that need indoor mapping. New students and visitors may find it difficult to navigate inside this building due to its multiple floors and classrooms. Currently, there is a floor plan located in this building, but it is not easy to find.

An IPS is a possible solution for mapping the Physical Sciences Building, but it can be costly and challenging to build and maintain. It is also not very scalable if the system were expanded to the entire campus, because it requires external hardware for each floor in each building. Hence, we turn to indoor mapping applications that can be accessed anywhere.

1.3 Objectives of the Study

The main objective of this study is to develop a web application that provides an indoor map of the UPLB Physical Sciences Building. Specifically, it aims to:

1. integrate point-to-point directions to aid users in finding the exact locations of classrooms, offices, and facilities within the UPLB Physical Sciences Building;

2. enable building administrators to generate a building's maps based on floor plans, allowing for updates and customization of the mapping content; and
3. assess the usability of the system through user testing to evaluate navigation accuracy, interface intuitiveness, and overall user satisfaction.

1.4 Significance of the Study

The web application developed in this study will be beneficial to visitors and new students who have classes inside the Physical Sciences Building. It will be easier for them to find the classrooms that they need to get to, and it will also save them time and prevent traffic in the corridors.

The results of this study may also serve as a basis for future indoor mapping implementations in UPLB. Aside from determining the feasibility of such a project, future studies will know what can be improved based on this study.

1.5 Scope and Limitations of the Study

The main focus of this research is to build a web application to navigate through the inside of the Physical Sciences Building located at the University of the Philippines Los Baños, Laguna. The end users of this research include individuals who are unfamiliar with the building, regular occupants of the building such as employees and students, and building administrators of the Physical Sciences Building. Using the application requires access to network connectivity, as the application may need to update map data and navigation information as it is being used, and it is also required when logging in.

The application does not track the user's real-time location; instead, users select their origin and destination, and the route is displayed on the map. Accessible routes and emergency exit routes are not yet mapped. The evaluation of the application was limited to

six student users and four building administrators from the University Planning and Maintenance Office (UPMO).

1.6 Definition of Terms

The following terms are used in this study as defined below.

Administrator. A user who logs in to the application and can create, edit, and publish floor layouts through the layout builder, in addition to using the navigation features.

Draft layout. An unpublished version of a floor layout that administrators can save, load, rename, and delete before publishing it.

Floor layout. The digital representation of a floor of the building, composed of rooms, hallways, stairs, bridges, and entrances positioned and sized on a two-dimensional plane.

Indoor positioning system. A system that approximates a user's location inside a building using devices installed in the building, such as Wi-Fi, infrared, Bluetooth, or computer vision.

Layout builder. The administrator feature used to create, edit, and publish the floor layouts on which navigation is based.

Location-based services. Technologies that use a device's location to provide services such as object positioning, navigation, and mapping.

Navigation graph. The network of nodes (rooms, doors, hallway segments, stairways, and the main entrance) and connecting edges that the application builds from the published layout and on which it computes routes.

Published layout. The finalized layout that serves as the reference for the map and routes shown to all users; also called the main layout.

Regular user. A student, employee, or visitor who uses the application to search for locations and view routes within the building.

Sandbox mode. A separate workspace in which administrators can explore the layout builder without affecting the main layout or the published map.

System Usability Scale. A ten-item questionnaire that produces a single usability score from 0 to 100.

Wayfinding. The process of determining and following a route from an origin to a destination.

Wing. A section of a floor of the Physical Sciences Building (Wing A, Wing B, or Wing C) that users can select when viewing the map.

CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Technological Trends in Mapping Applications

Location-based services (LBS) is an umbrella term for technologies that use a device's location to provide services. Object positioning, navigation, and mapping all count as LBS. Junglas and Watson [3] observed the difference between LBS and non-LBS systems through an empirical study involving navigation tasks, and it was found that individuals prefer LBS because it is efficient and easy to use. However, these individuals also had privacy concerns: they were willing to provide their location to network providers but hesitant to give the same information to other entities such as third-party apps.

Another technology that is closely related to LBS is the Geographic Information System (GIS). GIS is similar to LBS in that they both involve spatial and location data, but one main difference lies in their intended purpose: GIS is often used for mapping and analyzing geographic information, while LBS is more user-oriented and focuses on navigation and on adding that same functionality to everyday apps. GIS has made great progress over the past decades, but there are still areas for improvement, such as geographic modeling enhancement and human-oriented presentation of geographic information [4].

LBS and GIS are far from the only technological trends. Kunhoth et al. [5] provide an overview of various indoor positioning and wayfinding methods from 2014 to 2020. Among these technologies are radio frequency identification (RFID), ultra-wideband (UWB), and computer vision-based indoor mapping.

2.2 Commonly Used Navigation Apps

To understand how navigation technologies work, we first take a look at Google Maps, one of the most popular and feature-complete navigation apps. A report by Mehta et al. [6] analyzes this app in terms of its navigation and mapping. In terms of navigation, Google Maps uses the A* pathfinding algorithm. For its mapping, Google uses specialized devices such as a 360° camera attached to a car. This improves the accuracy of paths and provides 3D viewing through Street View. To calculate the estimated time of arrival (ETA), Google Maps collects data from its users on that particular route and approximates the traffic speed. This means that the more users there are, the more accurate the estimates will be. However, there are also ethical considerations here, and transparency is important so that users know whether their personal information is being used against them. Aside from Google Maps, an alternative mapping solution is OpenStreetMap, a community-driven digital map where volunteers can contribute to its development [7]. OpenStreetMap emphasizes collaboration, organizing "mapping parties" that annotate data on various geographical areas.

2.3 Campus Navigation Solutions

There exist several studies that involve indoor mapping for universities around the world. Vanjani et al. [8] implemented a navigation app for Jhulelal Institute of Technology (JIT) in Nagpur, India. The application was created to help college students reach JIT College from various outside places (inter-college), as well as offer directions to buildings and classrooms within the university (intra-college). For the inter-college stage, they used GPS and the Apple Maps API with a fixed destination towards JIT College. For the intra-college stage, the paths to and from each building were hard-coded, and a pre-made image

was shown on the screen. This method works, but it is not ideal since the number of possible paths is close to n^2 (where n is the number of buildings), so having a large number of buildings increases the manual work quadratically.

There are also several navigation studies in the Philippines. Castillo et al. [9] created a mobile navigation app for the Technological Institute of the Philippines (T.I.P.) Quezon City campus, intended for new students and visitors. First, the user is taken to a login screen. After creating an account and logging in, the user can choose a source and a destination building, and the app generates a path between them using a fuzzy logic algorithm. The advantage of this feature over [8] is that the paths are automatically generated. However, it may be prone to errors depending on the strength of the algorithm. As an added feature, the user can also see other active users (shown as pinpoints) within the same area using a channel selection algorithm through Bluetooth.

Similar to [9], Ramos et al. [10] created a locator map of Saint Michael's College of Laguna using Android Studio. It uses the A* algorithm for pathfinding. The layout of the campus map, including the buildings and the walkable roads, was manually created. The app also has a search bar where the user can search for a particular building, and it displays information about that building, including the classes one needs to take in that building. Overall, the app was rated "Excellent" based on 346 respondents.

A more gamified approach to campus navigation is found in the JUAN app of Don Mariano Marcos Memorial State University [11]. The mobile campus map incentivizes the user to go to different locations within the campus to collect BeeCoins, which represent the number of points the user has. Through this approach, the user can have fun while learning. The app used satellite view for the map layout and GPS for the user's location.

2.4 Indoor Navigation in Other Industries

Aside from campus navigation, indoor mapping can also be effective in other settings, such as finding stores in malls. A system proposed by Tulaskar et al. [12] uses Wi-Fi fingerprinting to locate the user and Dijkstra's algorithm for pathfinding. The application was developed using Android Studio and Node.js, and it can display directions from the user's location to other stores in the mall. However, it is unclear how multi-floor navigation functionality can be achieved based on this implementation.

Another implementation by Saeliw et al. [13] combines augmented reality (AR), geographic information systems (GIS), and GPS technologies to find stores. To do this, a database was kept storing data on the floor level, latitude, and longitude of each store. The user could then use their phone camera to locate the store through an AR popup. Again, it is unclear whether the floor level of the user was automatically determined or manually inputted.

Similar to [12], Tan et al. [14] use Wi-Fi instead of GPS to improve the accuracy of the user's indoor location. The mobile application used an existing indoor map found in Google Maps and provided an interface for locating stores inside Plaza Low Yat in Kuala Lumpur, Malaysia. On a scale of 1 to 5, users consistently rated it above 4 in terms of its usefulness, ease of use, efficiency, and aesthetics.

2.5 Challenges of Indoor Navigation

While indoor navigation is certainly beneficial to all kinds of users, it does not come without its challenges. A study by Zlatanova et al. [15] proposes a set of indoor mapping and modeling (IMM) issues that are worth researching. Among these problems are optimal navigation routing, emergency preparedness, and legal and security issues.

Indoor navigation is believed to promote accessibility, especially for users with mobility challenges. However, there is still a lack of actual testing with these individuals. According to an analysis of 111 existing wayfinding and navigation papers [16], persons with mobility issues had no involvement in the design and development of the navigation solutions in most papers. This makes it hard to prove the effectiveness of the product in real-life contexts.

Another problem with indoor navigation is its lack of accessibility standards and guidelines. Striegl et al. [17] report that in OpenStreetMap, on average, only one building per city has an indoor map, and among those buildings, the ones with accessibility information are even fewer.

One of the biggest challenges of indoor positioning, mapping, and navigation (IPMN) is its lack of standardization. Over the decades, multiple researchers and developers have tackled IPMN, but it is hard to undergo the process of industrialization because there is no clear set of rules to be followed. Gelernter and Maheshwari [18] also point out the lack of a standardized file format for indoor map data, making it harder to have a one-size-fits-all approach to speed up the indoor mapping process globally. To address this, Deng et al. [19] proposed a set of requirements for IPMN:

1. Indoor positioning can be classified into two types: requiring external equipment, or device-free. If using the former, the external equipment must be prepared in advance.
2. Indoor positioning must not be separated from map visualization, and map visualization should consider expressing symbols, color, and semantic information.

3. Indoor and outdoor positioning should be treated as separate modes in the application.
4. A consistency test is required for evaluating the accuracy and cost of positioning technologies.

CHAPTER III

METHODOLOGY

This chapter describes how the application was developed and evaluated. As shown in Figure 3.1, the study proceeded in five phases: requirements gathering, analysis and design, implementation, deployment, and user testing and evaluation.

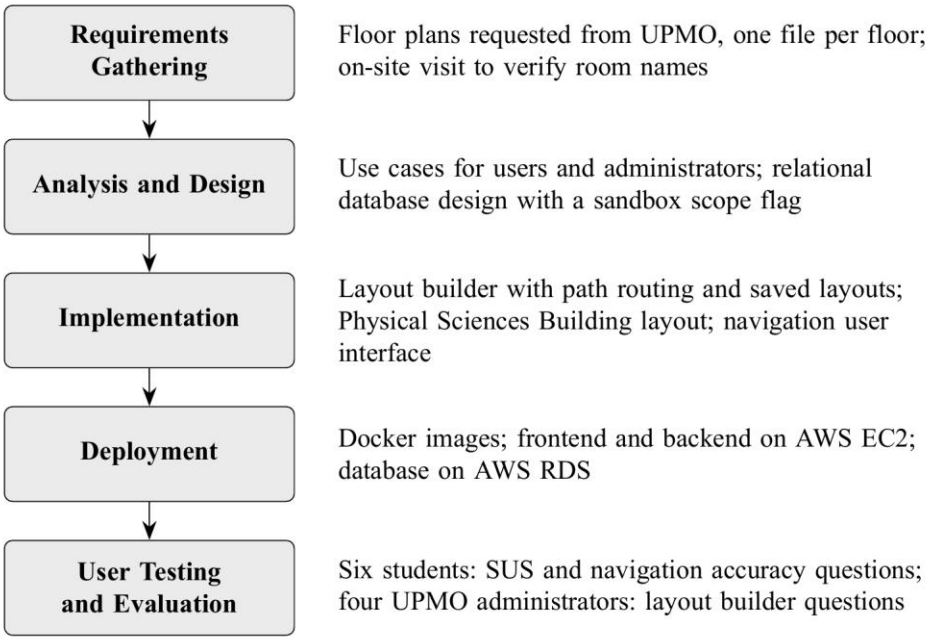


Figure 3.1. Phases of the study

3.1 Requirements Gathering

In order to retrieve the correct layout of the Physical Sciences Building, an email was sent to the University Planning and Maintenance Office (UPMO) to request the building's floor plans. A 2D layout of the building was provided, separated into three files, with each file mapping out an entire floor.

For the labeling of the rooms, a physical visit inside the building was conducted to cross-examine the accuracy of the floor plan against the actual rooms, including the name

of each room. After development of the application, another email was sent to UPMO to request user testing. Administrators from their office were asked to participate in testing the layout builder feature of the application.

3.2 Analysis and Design

Figure 3.2 shows the use-case diagram for the indoor navigation web application. Two actor roles were identified: users and administrators, each with distinct capabilities within the system. Users were able to search for rooms, select an origin and destination, and view the generated navigation route. Administrators, in addition to having access to navigation features, were able to create, edit, and publish floor layouts through the layout builder, giving them control over the data that powered the navigation feature for all users.

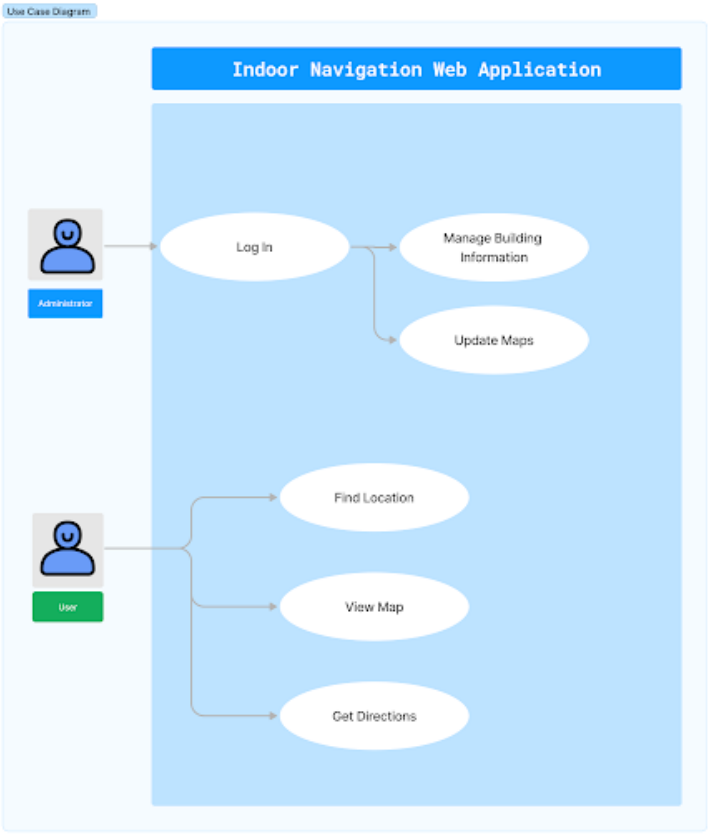


Figure 3.2. Use-case diagram of the indoor navigation web application

Based on these use cases, the system was designed around a relational database structure using MariaDB. During backend startup, the database service connected to the database and verified that the required schemas were present before the application became available. A scope flag was incorporated within the relevant tables to separate live, published records from sandbox records, allowing administrators to edit and test layout changes without affecting the version currently used for navigation.

Figure 3.3 shows the Entity Relationship Diagram (ERD) of the application database. The `login_accounts` table stores the administrator accounts used to access the layout builder. The `draft_map_layouts` table stores unpublished layout drafts, and each draft may be published as entries in the `map_layouts` table, which holds the published layouts together with a flag marking the current version. Each published layout may in turn be cached by entries in the `navigation_graph_caches` table, which stores the navigation graph computed from that layout. The buildings, floors, rooms, and associated navigation data are held in the `layout_json` and `graph_json` columns, while the `is_sandbox` column serves as the scope flag that separates sandbox records from live records.

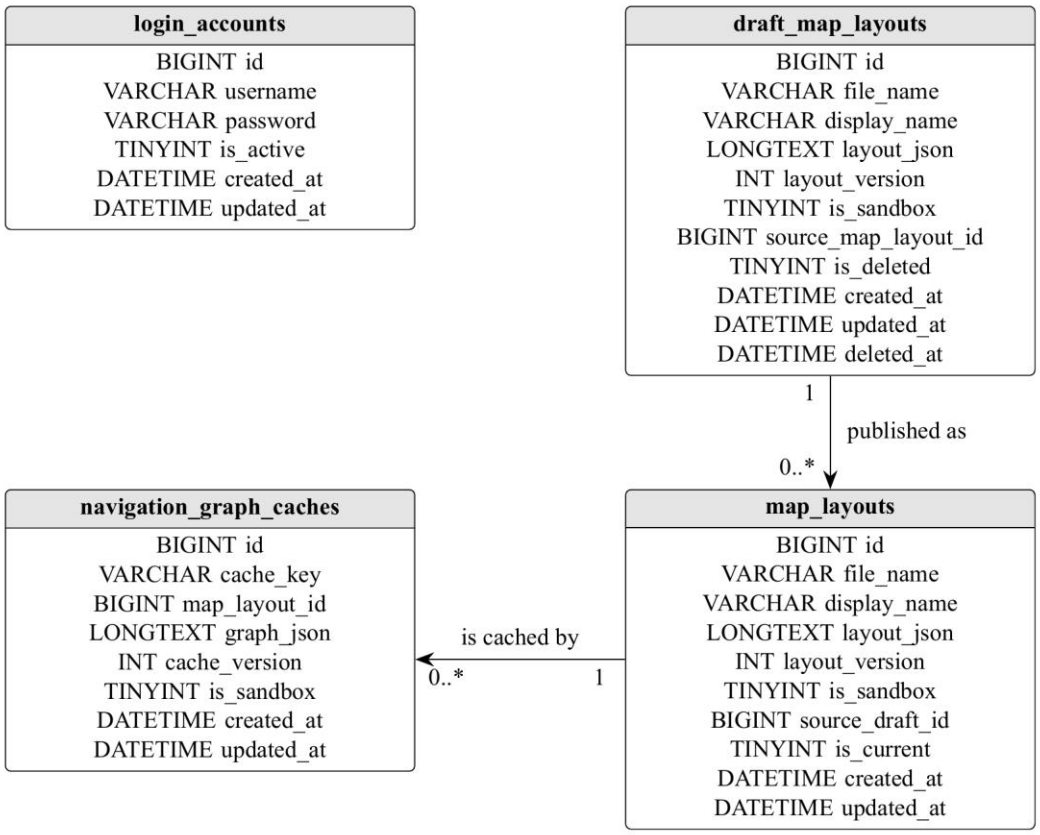


Figure 3.3. Entity relationship diagram of the application database

3.3 Development Tools

A laptop with the specifications listed in Table 3.1 was used to develop the indoor navigation web application.

Table 3.1. Hardware specifications of the development machine

Component	Specification
Operating system	Windows 11, 64-bit
Processor	Intel® Core™ i5-8250U, 1.6 GHz
Graphics card	NVIDIA GeForce GTX 1050
Memory	20 GB DDR4, 2800 MHz

The software development tools listed in Table 3.2 were used in the development of the application.

Table 3.2. Software development tools

Tool	Description
Python	An interpreted, object-oriented, high-level programming language, used to build the backend component of the application
MariaDB	An open-source relational database management system
React	An open-source front-end JavaScript library, used to build the frontend component of the application
Visual Studio Code	A source-code editor used for developing the application

3.4 Implementation

The application was built first with the layout builder features in mind. The goal was to create a working map layout with the integration of path routing and a persistence state to save layouts. A layout of the Physical Sciences Building was then created using the layout builder, guided by the floor plan provided by UPMO. After that, focus shifted to the user interface, allowing users to select between different rooms. Finally, once the application was built, it was deployed on AWS to test the application's stability outside the local development environment.

3.5 Deployment

The application was deployed on Amazon Web Services (AWS). The following AWS services were used:

- **Elastic Compute Cloud (EC2)** – an EC2 instance was created to configure and run both the frontend and the backend services.
- **Relational Database Service (RDS)** – a managed cloud database that handled the storage of data and database operations.

The application's backend and frontend were each packaged as separate Docker images. This containerization approach supported a more consistent and repeatable

deployment process across the development lifecycle, since the same image built and tested locally could be deployed to the cloud without modification. Both containers run on the same AWS EC2 instance and communicate through a shared Docker network. The frontend container serves the built React application through an nginx web server, which makes the application publicly accessible via a web URL and forwards the application's API requests to the backend container. The backend container runs the Python service, which processes these requests and connects to an AWS RDS instance that is responsible for persisting and retrieving application data. Figure 3.4 summarizes this deployment architecture.

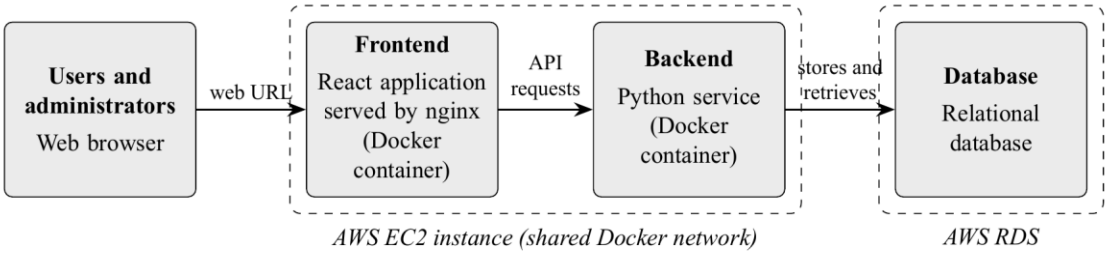


Figure 3.4. Deployment architecture of the application

3.6 User Testing and Evaluation

3.6.1 Participants

User testing was conducted with two participant groups: regular users and building administrators. A total of ten (10) respondents took part: four (4) building administrators from the University Planning and Maintenance Office (UPMO) and six (6) students of the University of the Philippines Los Baños.

3.6.2 Procedure

For building administrators, the focus of testing was the map layout builder feature, specifically whether the tools for the map layouts functioned correctly and were practical to use. They logged in to the system, accessed the layout builder, and tested the available

editing functions, such as adding, modifying, saving, and publishing map elements. They then verified whether the published changes were correctly reflected in the navigation interface.

For regular users, the focus of testing was navigation accuracy, particularly whether the application could correctly guide them from a selected origin to a specified destination within the building and whether the displayed route matched the actual path.

A live demonstration of the application was conducted online via Zoom, with separate sessions held for regular users and building administrators. Questions related to application features were accommodated during this period, and respondents were then asked to answer a Google Form containing questions related to system usability and navigation accuracy. A one-week testing period was given so that all users would have ample time to explore and use the features of the application.

3.6.3 Evaluation Instruments

To quantify the usability of the web application, the System Usability Scale (SUS) [20] was used. This is a reliable tool for measuring the usability of various systems and services. The SUS provides a set of ten statements (Appendix A), each rated on a five-point Likert scale ranging from “Strongly disagree” (1) to “Strongly agree” (5). This scale is designed to evaluate a broad range of interactive products by deriving a single usability score that reflects a composite measure of the overall effectiveness, efficiency, and satisfaction with the system.

Following the standard SUS scoring procedure [20], each odd-numbered (positively worded) statement contributes its response minus one, and each even-numbered

(negatively worded) statement contributes five minus its response. The ten contributions are summed and multiplied by 2.5, which yields a score from 0 to 100:

$$SUS = 2.5\left[\sum_{i\in\{1,3,5,7,9\}}(x_i - 1) + \sum_{i\in\{2,4,6,8,10\}}(5 - x_i)\right] \quad (3.1)$$

where x_i is the response to statement i. The resulting scores were interpreted using the Sauro–Lewis curved grading scale [21], which ranks a SUS score against the scores of hundreds of other systems, together with the adjective ratings of Bangor et al. [22], as combined by Sauro [23]. Table 3.3 shows the resulting interpretation, in which the average SUS score of 68 corresponds to a grade of C.

Table 3.3. Interpretation of SUS scores

SUS score	Percentile range	Adjective rating
84.1–100.0	96–100	Best Imaginable
80.8–84.0	90–95	Excellent
71.1–80.7	60–89	Good
51.7–71.0	15–59	OK
25.1–51.6	2–14	Poor
0.0–25.0	0–1.9	Worst Imaginable

Source: Sauro [23]; grades and percentile ranges from the Sauro–Lewis curved grading scale [21].

To evaluate navigation accuracy, a qualitative testing phase was conducted. After completing the assigned navigation tasks, the regular users answered a set of six open-ended questions (Appendix B) based on their experience using the application. The responses were analyzed to determine whether the application was capable of providing accurate and understandable indoor navigation guidance. The building administrators answered a separate set of seven open-ended questions about the layout builder (Appendix C).

CHAPTER IV

RESULTS AND DISCUSSION

4.1 Application Features

4.1.1 Location Search

The application enables users to search for specific destinations within the building by providing an origin and a destination. This feature serves as the primary entry point for route generation. Figure 4.1 shows the location search feature.



Figure 4.1. Location search

4.1.2 Multiple Floor Support

The application supports the viewing of buildings with multiple floors, allows users to navigate between the different floor levels, and provides the option to select different wings for each floor. Figure 4.2 displays the options for the user to select from specific areas of the building.

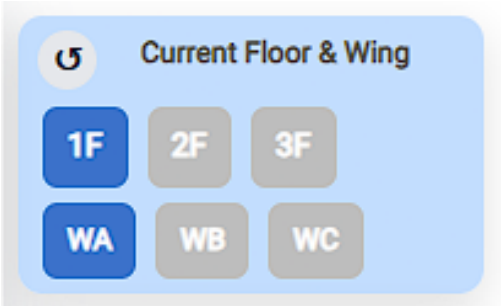


Figure 4.2. Floor and wing selector

4.1.3 Map Visualization

The application provides an interactive indoor map that allows users to zoom, pan, and reset the viewing area. Figures 4.3 and 4.4 show the 2D floor view and the 3D visualization when navigating.

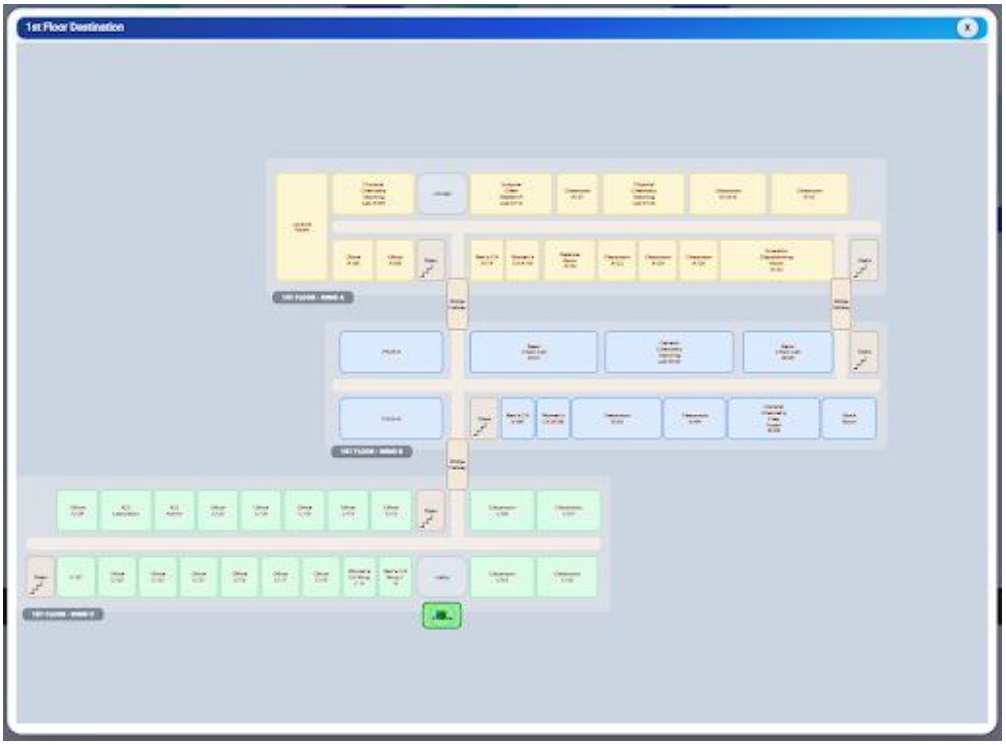


Figure 4.3. Two-dimensional (2D) map view

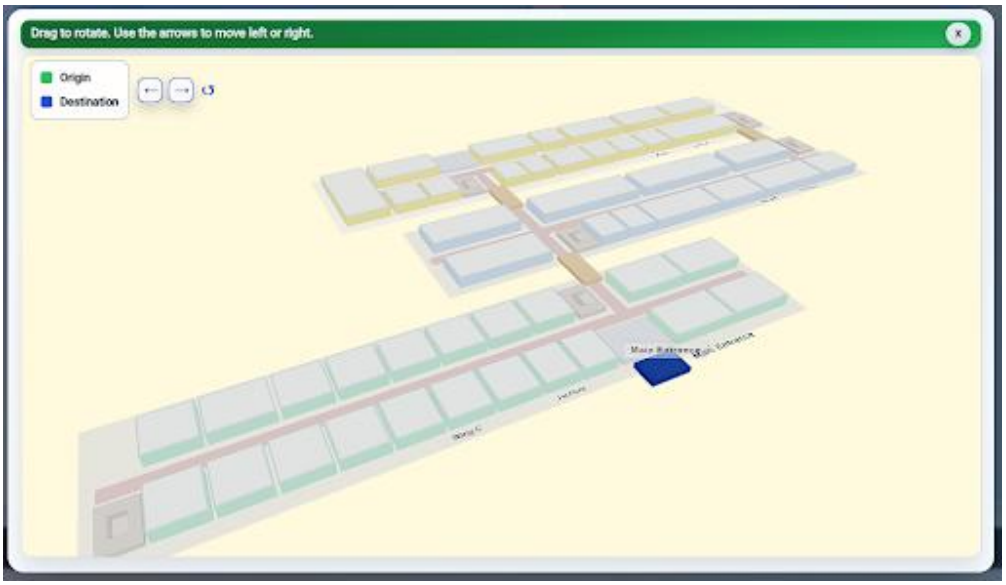


Figure 4.4. Three-dimensional (3D) map view

4.1.4 Multi-Floor Layout Builder

The application includes a layout builder that enables administrators to create and edit an indoor map to be seen by the users. Figure 4.5 shows the panel for the layout builder, which includes the options available to the administrator.



Figure 4.5. Layout builder panel

4.1.5 Layout Builder Map Drafts

The application allows administrators to save, load, rename, and delete draft map versions. Figures 4.6 and 4.7 show the Map Layouts and Map Drafts modals, which allow the administrator to select between map versions.

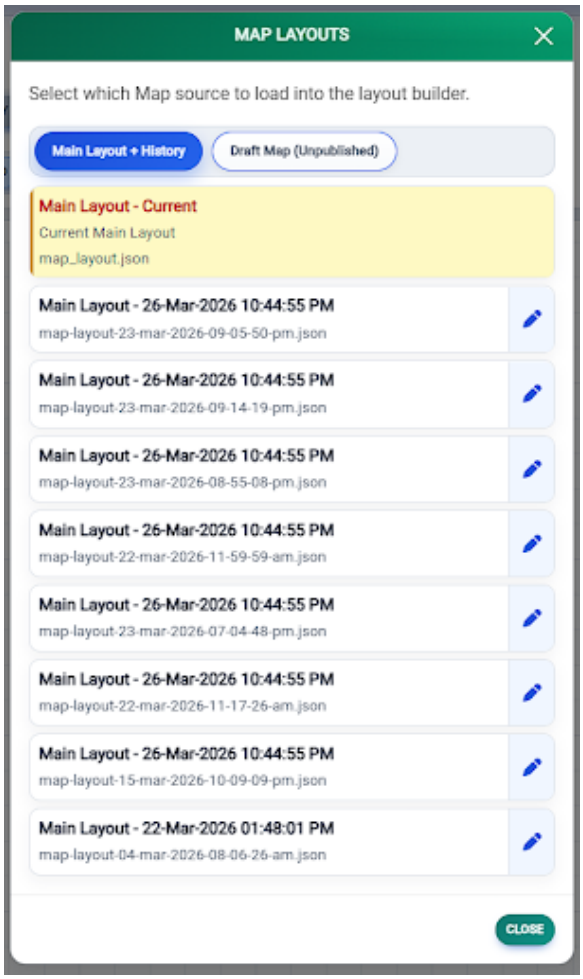


Figure 4.6. Main map layouts

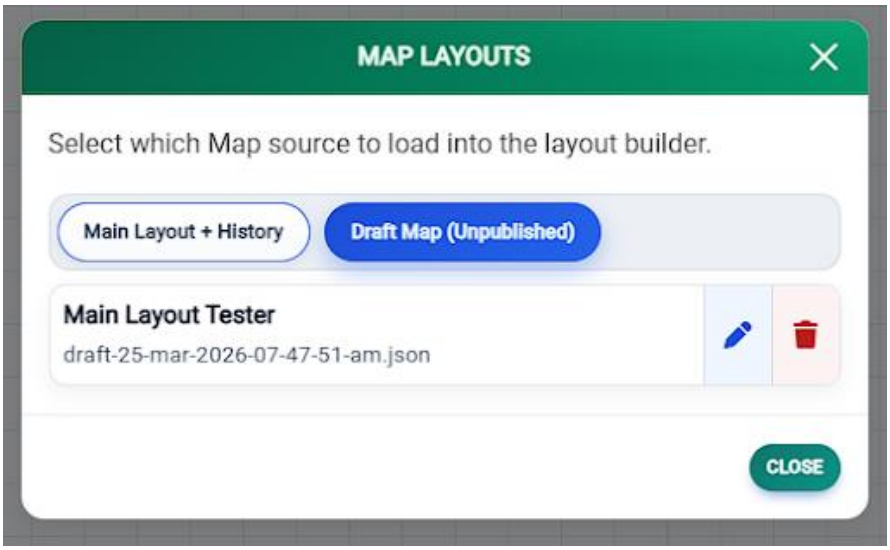


Figure 4.7. Map draft layouts

4.1.6 Sandbox Mode

The application includes a sandbox mode in which administrators can explore the features of the map layout builder without the risk of overwriting the main map edits and the published map being displayed to the users. Figure 4.8 shows the sandbox mode.

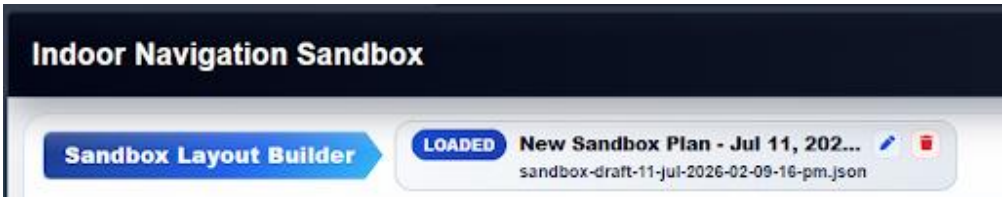


Figure 4.8. Sandbox mode

4.1.7 Auto-Generated Layouts

The application has an auto-generate map feature that allows the administrator to select a set of predefined layouts to assist them with map creation, as shown in Figure 4.9.

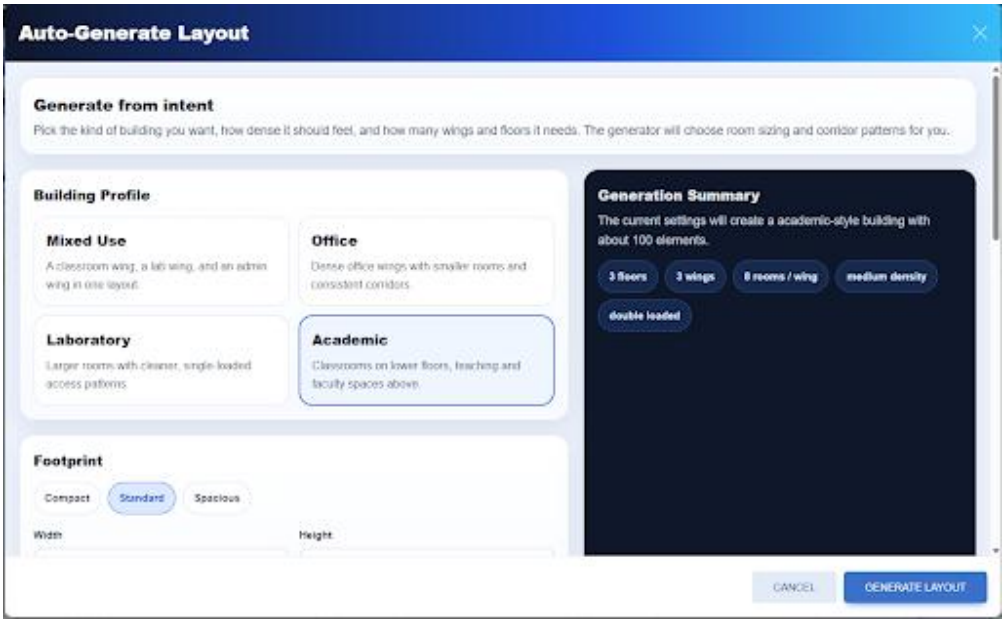


Figure 4.9. Auto-generate map layout

4.2 Types of Users

The application recognizes two types of users:

Admin. Administrators can modify the mapping of the building, manage location data, and update floor plans.

User. Users can search for locations within the building and use the navigation features to access floors, rooms, and routes.

4.3 Mapping of the Floor Layout

The floor layout of the building is developed through the application's layout builder, where the admin creates and organizes the map structure for each floor. The admin directly encodes the building layout within the web application.

Using the layout builder, the admin defines the major spaces of the building, such as rooms, hallways, stairs, bridges, and entrances. In this process, each floor is treated as a two-dimensional plane with x and y coordinates, allowing the admin to position each component based on its x and y location and to size it according to its width and height within the floor plane. Labels are assigned so that the digital floor plan reflects the actual arrangement of the building.

The admin can revise the floor plan during the mapping process, and draft versions of the layout can be created and edited before publication. Once the admin finalizes the layout, the floor plan is published and becomes the reference used for navigation display within the application. Previously published versions of a map can also be reverted to and saved as a new draft, allowing the admin to roll back changes when needed.

After the map is published, the application uses the mapped structure and computes the navigation routing based on the details provided by each component to determine the path from one point to another, as described in the next section.

4.4 Route Computation

PathFinder computes routes with the A* search algorithm, the same algorithm used by Google Maps [6] and by the campus locator map of Ramos et al. [10]. The search runs on a navigation graph that the backend builds from the published layout.

Navigation graph. Each room, door, hallway segment, stairway, and the main entrance becomes a node positioned at its x and y coordinates on its floor. Edges join consecutive hallway segments along each corridor; connect each room, through a door node placed on the room wall facing the corridor, to the nearest suitable hallway on the same floor; connect each stairway to a hallway on its floor and to the stairways directly above and below it; and link the wings through the bridges. An edge is not added when the straight segment between its two nodes would cut through the interior of another room. Because building the graph is computationally expensive, the result is stored in the `navigation_graph_caches` table and reused until a different layout is published.

Edge cost. The cost of moving between two connected nodes u and v is

$$c(u, v) = d(u, v) + 150|f_u - f_v| + p_J(u, v) + p_D(u, v) \quad (4.1)$$

where $d(u, v)$ is the straight-line distance between the two nodes on the floor plane, in layout coordinate units (for a hallway, measured to the nearest point of its rectangle); f_u and f_v are the floor numbers of the two nodes; $p_J(u, v)$ is 120 when both nodes are room-connection junctions, 24 when only one of them is, and 0 otherwise; and $p_D(u, v)$ is 25 when both nodes are doors and 0 otherwise. The floor term discourages unnecessary floor changes, while the junction and door terms keep routes on the main hallways instead of passing from one room entrance to the next.

Search. A* repeatedly expands the unexplored node with the lowest estimated total cost

$$f(n) = g(n) + h(n) \tag{4.2}$$

where $g(n)$ is the lowest known cost from the origin to node n computed with Equation 4.1, and the heuristic $h(n)$ is the straight-line distance from n to the destination. The search skips any edge that crosses the interior of a room not marked as passable, so a route never cuts through a room.

Routes between wings. When the origin and destination are in different wings, the route must cross one of the bridges that connect Wings A, B, and C. The backend prepares candidate sequences of bridge waypoints, computes each leg between consecutive waypoints with the same search but with the heuristic set to zero, which makes it equivalent to Dijkstra's algorithm, joins the legs, and keeps the candidate whose drawn route is shortest. When the origin and destination are on the same floor but the best route leaves that floor, the route is recomputed with the nodes on other floors excluded. If a route passes through the door of a room other than the origin or destination, it is recomputed with that door excluded.

Finally, the route is simplified for display and returned to the frontend as one Scalable Vector Graphics (SVG) path per floor, which the map view draws as the route overlay.

4.5 Navigation User Interface

Users can find specific locations within the building with the search functionality and select their origin and destination points by typing in the search bars or by choosing the rooms in the room list selector. Figures 4.10 and 4.11 show the user interface from the

user's perspective, whereas Figure 4.12 displays the interface from the administrator's perspective.



Figure 4.10. Main user interface

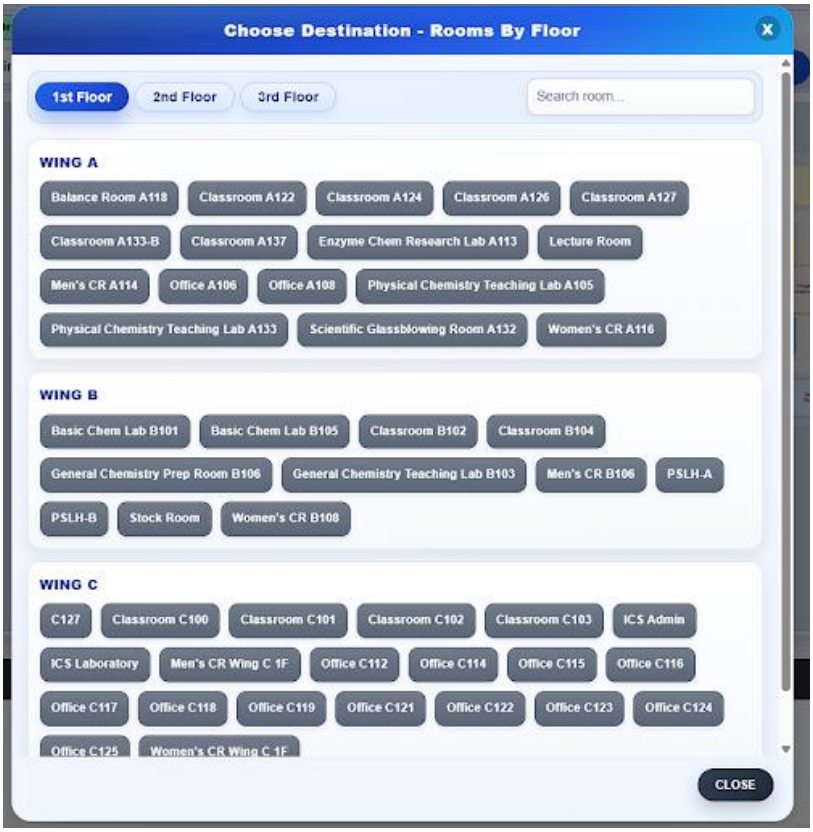


Figure 4.11. Room selector

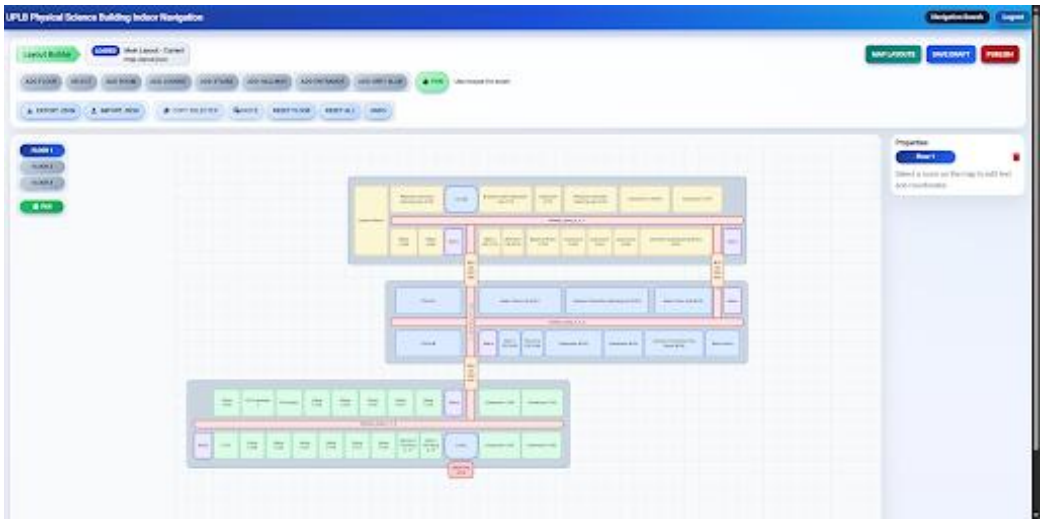


Figure 4.12. Administrator user interface

4.6 Non-Functional Requirements

Usability. This requirement was met through a guided user interface. The application provides assisted search suggestions, room-list browsing, clear map controls, visible route overlays, and consistent navigation elements, making it easier for users to understand and operate the system.

Performance. This requirement was met by having the application implemented to be browser-based, which means it can operate and is compatible on devices capable of running modern web browsers without requiring a platform-specific installation

Scalability. This requirement was met through the separation of frontend and backend services, containerized deployment, graph caching, and a database for storage persistence to accommodate larger map layouts and multiple maps.

Security. This requirement was met through administrator login, protected access to the layout builder only for administrators, session expiration, and controlled Cross-Origin Resource Sharing (CORS) settings. However, more robust authorization would further improve security.

Availability. This requirement was supported through the application's AWS-based deployment and service health mechanisms. The system was deployed on an AWS EC2 instance using Docker containers for both the frontend and backend, allowing the services to remain isolated and easier to restart when needed.

4.7 User Testing Results

A total of ten (10) responses were collected, with four coming from building administrators from UPMO and six from students studying at the University of the Philippines Los Baños.

4.7.1 System Usability Scale

Table 4.1 shows the raw questionnaire results from the users, and Table 4.2 shows the SUS score of each respondent and the rating to which the score belongs. The computation of each score using Equation 3.1 is shown in Appendix D.

Table 4.1. Raw SUS responses per respondent

Respondent	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
R1	4	1	5	1	5	3	5	5	5	5
R2	3	1	5	1	4	3	4	1	5	1
R3	4	2	4	2	5	3	5	4	4	4
R4	5	1	5	1	5	1	5	1	5	2
R5	5	1	5	1	5	1	5	1	5	1
R6	3	1	4	3	4	1	4	2	4	5

Responses range from 1 (Strongly disagree) to 5 (Strongly agree).

Table 4.2. SUS score, grade, and rating per respondent

Respondent	SUS Score	Rating
R1	72.5	Good
R2	85.0	Excellent
R3	67.5	OK
R4	97.5	Excellent
R5	100.0	Excellent
R6	67.5	OK
Mean	81.67	Excellent
Median	78.75	Good

Grades and ratings follow Table 3.3.

The six end users who completed the system usability questionnaire produced individual scores of 72.5, 85.0, 67.5, 97.5, 100.0, and 67.5, for a mean of 81.67 and a median of 78.75. The mean score of 81.67 falls within the "Excellent" range of the scale in Table 3.3, indicating strong overall satisfaction with the navigation experience. Three respondents gave scores in the "Excellent" range, while two respondents scored at the lower end (67.5, rated "OK"), suggesting that while the majority found the system highly usable, a subset encountered enough friction to rate the experience more moderately.

4.7.2 Navigation Accuracy

Qualitatively, all six student respondents confirmed that the application successfully led them to their intended destination and described the routes as accurate or mostly accurate. Two respondents noted that routes involving stairways were occasionally not the shortest or most efficient option available, and another respondent recommended that the system be tested against a wider variety of stairway configurations. Respondents highlighted the three-dimensional view as an effective way to visualize paths across

multiple floors compared to the two-dimensional view. Clarity in routing was rated favorably, with respondents describing the interface as straightforward and simple to understand.

4.7.3 Layout Builder Evaluation

The four building administrators who tested the layout builder feature described the tool as generally intuitive and responsive, and useful for early-stage planning of floor layouts. Respondents pointed out that the interface was straightforward to navigate and use, and that the resulting layouts helped communicate proposed arrangements to clients. One respondent noted that the application functioned effectively like a bubble diagram, a tool used for visualizing room connections before a final layout is committed. One feature identified for improvement was the addition of an autosave function; respondents mentioned that accidental misclicks could cause unsaved progress to be lost. Administrator respondents also pointed out that they had difficulty distinguishing the purpose of the Layout Builder from the Sandbox mode, which added complexity to understanding the separation between the two functions.

CHAPTER V

CONCLUSION AND FUTURE WORK

5.1 Conclusion

Taken together, the two testing groups suggested that the application performed strongly on its core function, indoor navigation, while leaving room for refinement in the layout builder. Users rated the navigation experience highly, with a mean score falling within the "Excellent" range. Building administrators, on the other hand, found the layout builder and sandbox mode useful for building layout planning, helping communicate layout proposals to stakeholders and clients. Some limitations of the application evaluation include the small sample size of both testing groups, consisting of six end users and four administrators. This limits how confidently these results can be generalized to a wider range of users. Additionally, testing was limited to a one-week period for both types of participants, which does not capture how usability holds up with repeated or long-term use of the application. This is particularly relevant for building administrators, who may create and manage large, multi-story buildings over time.

5.2 Future Work

For future work, a primary area of focus is expanding the system beyond its current scope to cover other buildings across the campus, allowing users to navigate between buildings rather than within a single building alone. To complement this expansion, the development of native iOS and Android applications is recommended, as a transition from a web-based platform to dedicated mobile applications could offer an improved overall user experience.

The application should also integrate accessibility features, which involve mapping and displaying accessible routes, such as paths incorporating PWD (persons with disability) ramps, elevators, and accessible restrooms. Relatedly, integrating clear displays of emergency exit routes is recommended to enhance the safety and preparedness of all users within the building.

With respect to the navigation feature specifically, future versions should support real-time location tracking to guide users continuously as they move toward their destination, rather than relying on a static route display, and should display doorways along the suggested path to further improve route clarity. Further routing optimization and testing are also recommended to broaden the scope of navigation scenarios evaluated, including more varied stairway configurations and larger, multi-floor paths.

On the administrator side, the layout builder would benefit from several refinements identified during testing. These include an autosave feature to prevent the loss of in-progress map drafts, a clearer distinction between the Layout Builder and Sandbox features to reduce confusion between the two, and support for real-world unit scaling with gridlines or rulers to improve drawing precision. The ability to upload a 2D or CAD reference plan to serve as a guide when constructing more accurate layouts is also recommended. Improving the responsiveness and synchronization of the layout builder's editing tools would further support administrators working on larger, more complex building layouts.

To support long-term map accuracy, a crowdsourced update and feedback system is recommended, through which users can report inaccuracies or suggest corrections

directly within the application, helping keep building layouts current without relying solely on administrator updates.

Finally, integrating the system with outdoor campus navigation is recommended to provide a connected wayfinding experience for users moving between buildings.

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APPENDICES

APPENDIX A

SYSTEM USABILITY SCALE QUESTIONNAIRE

Respondents rated each statement from 1 (Strongly disagree) to 5 (Strongly agree).

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

APPENDIX B

NAVIGATION ACCURACY QUESTIONS

The following open-ended questions were answered by the regular users.

1. Did the application lead you to the correct destination? Describe what happened during the navigation process.
2. Was the route provided by the application accurate? Explain why or why not.
3. How easy was it to understand the route displayed on the map?
4. Were there any parts of the route that were confusing, unclear, or incorrect? If yes, describe them.
5. Did you encounter any mismatch between the route shown by the application and the actual path that should be taken inside the building? If yes, describe the mismatch.
6. What suggestions can you provide to improve the navigation accuracy of the application?

APPENDIX C

LAYOUT BUILDER EVALUATION QUESTIONS

The following open-ended questions were answered by the building administrators.

1. Which parts of the layout builder felt easy to understand, and which parts felt unnecessarily complex?
2. Can you describe your experience creating or editing a building layout using the system?
3. How well did the different features of the layout builder work together while you were making changes?
4. Were there any parts of the layout builder that felt confusing or difficult to predict? Please explain.
5. How confident did you feel that the layout you created or edited was correct and usable for navigation?
6. Would you like to use this application in the future?
7. What suggestions do you have to improve the layout builder so administrators can create more accurate and reliable maps?

APPENDIX D

SUS SCORE COMPUTATION

Table D.1 shows how each respondent's SUS score was obtained from the raw responses in Table 4.1 using Equation 3.1.

Table D.1. SUS score computation per respondent

Respondent	Odd items $\Sigma(x_i - 1)$	Even items $\Sigma(5 - x_i)$	Total	SUS Score (Total × 2.5)
R1	19	10	29	72.5
R2	16	18	34	85.0
R3	17	10	27	67.5
R4	20	19	39	97.5
R5	20	20	40	100.0
R6	14	13	27	67.5